

# Differentially Private Preference Data Synthesis for Large Language Model Alignment

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## Abstract

Preference alignment is a central post-training step for large language models (LLMs), and its empirical value depends on pairwise human comparisons that often contain sensitive prompts, responses, and judgments. We propose **DPPrefSyn**, an algorithm that converts private preference data into differentially private (DP) synthetic preference data for privacy-preserving alignment. DP-PrefSyn is grounded in the Bradley–Terry preference model and the geometry of pairwise preference differences: it learns DP-protected preference structure from private data, uses public prompts to synthesize comparison pairs, captures heterogeneous annotator preferences through per-cluster linear reward models, and applies DP Principal Component Analysis (DP-PCA) to improve sample efficiency. Across question-answering and summarization benchmarks, the empirical pattern is consistent: DPPrefSyn delivers strong win rates under strict DP budgets and often exceeds both DP fine-tuning baselines and non-private fine-tuning on the original preference data. For example, at  $\epsilon = 2$  on Anthropic-HH with Pythia-2.8B and DPO, DPPrefSyn reaches a GPT-4o win rate of 56.48% versus 37.02% for DP-FT on original data. These comparisons show that DP synthetic preference data can be a practical, reusable, and quantitatively competitive alternative for LLM alignment. To the best of our knowledge, this is the first work to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

## 1. Introduction

Preference alignment methods such as RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024) have become

standard tools for improving LLM behavior against human expectations. Their empirical signal is pairwise: for each prompt, annotators compare two candidate responses and identify the preferred one. Alignment algorithms then use these comparisons to increase the model’s relative likelihood of preferred responses over dispreferred ones (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). Because these pairwise comparisons have repeatedly translated into practical gains, preference alignment is now used in chat assistants (Achiam et al., 2023), mathematical reasoning systems (Shao et al., 2024), and code generation models (Shen et al., 2023).

The same data that makes alignment effective also creates a privacy bottleneck. Real preference datasets can contain user prompts about health, identity, or other sensitive topics, while the associated feedback can expose beliefs, preferences, and behavioral patterns (Li et al., 2023; Yu et al., 2024). Differential privacy (DP) (Dwork et al., 2006b) provides a rigorous framework for controlling these risks, and prior DP alignment methods report encouraging privacy-preserving results (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a). Yet the comparison structure remains incomplete: some methods privatize prompts (Yu et al., 2024), others protect labels (Chowdhury et al., 2024; Zhang et al., 2025), and many are tied to a specific training algorithm such as RLHF (Wu et al., 2023a). These choices leave unprotected fields or limit reuse across newer objectives such as DPO. They also keep utility tied to the size and quality of costly private annotations, which can weaken empirical alignment performance when private data are scarce. These gaps lead to the central question of this work:

*Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining compatibility with emerging alignment methodologies?*

In this work, we answer this question affirmatively and frame the contribution around the resulting privacy–utility comparisons:

- We introduce DPPrefSyn (Figure 1), a DP synthetic preference-data generator designed so that each empirical comparison preserves the pairwise structure used by downstream alignment. The method combines three com-

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ponents: 1) *Capturing diverse human preferences via shared linear structure of rewards and clustering.* The Bradley–Terry model and a linear reward structure let us compare prompt–preferred and prompt–dispreferred embedding differences, cluster samples with similar directions, and fit a shared reward model within each cluster. 2) *Improving learning efficiency through DP-PCA.* DP-PCA projects high-dimensional preference differences into a compact subspace, improving clustering stability and reward learning under the same privacy budget. 3) *Saving privacy budgets by using public prompts.* Public prompts provide the query side of synthetic examples, while DP-trained reward models determine preferred and dispreferred responses; this concentrates privacy spending on the private preference signal rather than on prompt generation.

- We give a privacy analysis showing that DPPrefSyn satisfies  $(\epsilon, \delta)$ -DP under an explicit budget decomposition. The accounting allocates  $\epsilon_0$  to DP-PCA (Amin et al., 2019),  $\epsilon_1$  to DP-KMeans (Su et al., 2016), and the remaining budget to per-cluster DP-SGD reward training. The Privacy Random Variable (PRV) accountant (Gopi et al., 2021) composes the DP-SGD cost, and post-processing ensures that the released synthetic dataset can be reused across alignment algorithms and LLM backbones without additional privacy loss.
- We evaluate DPPrefSyn on OpenAssistant (Köpf et al., 2023), Anthropic-HH (Bai et al., 2022), and TL;DR summarization (Stiennon et al., 2020), reporting method-by-method win-rate comparisons across privacy budgets, alignment objectives, model families, and public-data scales. The margins are favorable and consistent: with  $\epsilon = 2$ , Pythia-2.8B trained by DPO on DPPrefSyn data reaches a GPT-4o win rate of 56.48% on Anthropic-HH, compared with 37.02% for DP-FT on original data. The experiments further show that the same DP synthetic dataset can support multiple preference optimization methods, including DPO and RLHF-style training, while maintaining strong privacy constraints.

## 2. Related Work

**Differentially Private Alignment of LLMs.** Existing DP alignment methods provide important baselines but protect different units and therefore induce different utility comparisons. Wu et al. (2023a) adapt PPO to the DP setting, giving privacy protection for reinforcement-learning alignment but remaining *limited to PPO*. Chowdhury et al. (2024) study reward estimation from preference feedback under label-DP (Ghazi et al., 2021), and Zhang et al. (2025) propose AUP-RLHF as a user-level label-DP framework for RLHF. These methods protect preference labels, but they *do not address the sensitivity of prompts or responses themselves*;

consequently, their privacy guarantees do not cover all fields used in pairwise alignment examples. They also depend on the available private preference annotations, which can limit utility when annotation volume is small. Yu et al. (2024) instead protect sensitive user *instructions* by generating DP synthetic *instructions* before annotation and fine-tuning.

A complementary set of baselines studies private LLM fine-tuning on general labeled data rather than pairwise comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022). Another line protects in-context prompts for in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b). These works motivate our evaluation emphasis on the preference-specific setting, where the unit of comparison is a prompt with two ranked responses.

**Differentially Private Synthetic Text Generation.** DPPrefSyn also relates to DP synthetic text generation, where the central quantitative trade-off is whether synthetic data can retain downstream utility while satisfying a formal privacy budget. One line first fine-tunes a language model on private data under DP and then samples synthetic text from that model (Yue et al., 2023; Mattern et al., 2022; Miresghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For example, Yu et al. (2024) privately fine-tune LLaMA on private instruction data and then use the resulting model to generate synthetic text for supervised instruction tuning.

Because DP fine-tuning can be expensive or unavailable for closed-source models, another line uses LLM APIs to generate candidate synthetic data and privately select examples close to the private distribution (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). For instance, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples using embedding-space similarity to private data. *Our work follows this synthetic-data perspective but changes the object being preserved: rather than matching single texts, DPPrefSyn preserves pairwise preference comparisons and evaluates whether those comparisons improve downstream alignment win rates.*

**Diversity in Human Preferences.** Human preferences are not a single homogeneous signal (Denton et al., 2021; Aroyo et al., 2023a;b; Chakraborty et al., 2024). Sociodemographic differences, personal beliefs and biases, domain expertise, and ambiguity in language can all change which response is judged better (Sandri et al., 2023; Vogels, 2021). This motivates personalized or group-aware alignment methods (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025); for example, Singh et al. (2025) use meta-learning so an LLM can adapt to individual users from a small number of preference examples. DPPrefSyn incorporates this heterogeneity into the synthetic-data pipeline by clustering preference-

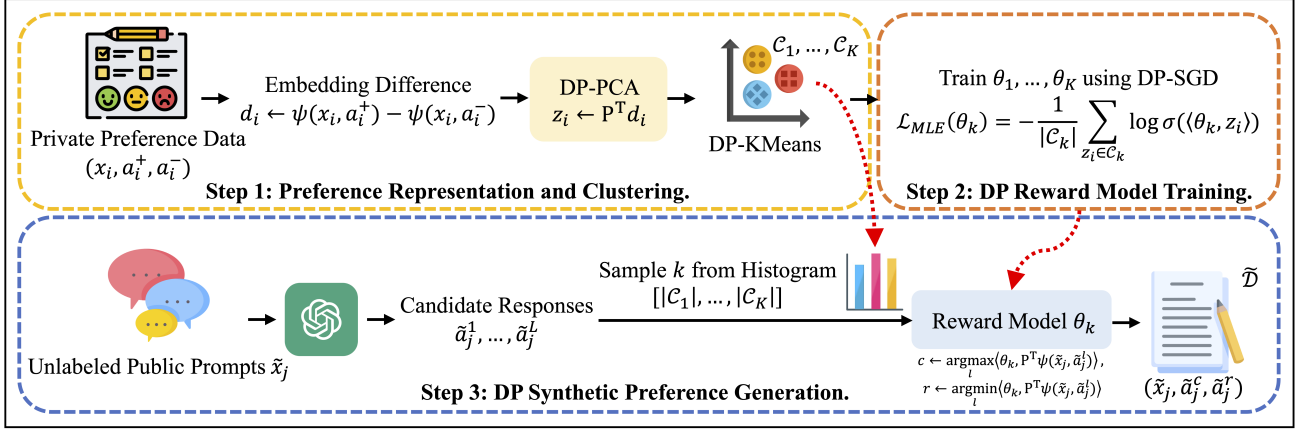


Figure 1. Overview of DPPrefSyn. The pipeline preserves the quantitative comparison structure of private preference data in 3 DP-accounted stages: 1) convert each private pair into an embedding-difference vector and cluster these vectors with DP-PCA and DP-KMeans, 2) train one DP reward model per cluster using DP-SGD, and 3) synthesize public-prompt preference pairs by sampling from the DP-protected cluster distribution and selecting high- versus low-reward candidate responses.

difference vectors, and the ablations in Section 5.3 quantify the utility loss when this structure is removed.

### 3. Problem Definition

**Preference Alignment of LLMs.** We study the standard pairwise-feedback setting, where the empirical unit is a ranked comparison rather than a single labeled response. The preference dataset is  $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ : each example contains a prompt  $x_i$

Following RLHF and DPO analyses (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we use the Bradley–Terry model (Bradley & Terry, 1952) to turn each observed comparison into a probabilistic margin between latent rewards:

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} = \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)),$$

where  $a_i^+ \succ a_i^-$  denotes the observed preference for  $a_i^+$  over  $a_i^-$ ,  $r^*(x, a)$  is the latent reward function, and  $\sigma(z) = 1/(1 + e^{-z})$  is the sigmoid function mapping the reward difference to a win probability.

This probabilistic comparison is the common substrate for RLHF, which explicitly estimates a reward model, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy.

**Differentially Private Preference Data Synthesis.** The private object to protect is the full comparison dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , including prompts, responses, and preference labels. The adversary may observe outputs of an aligned LLM and attempt to infer whether particular private examples or attributes were used. DP-

PrefSyn therefore releases a synthetic comparison dataset  $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$  that preserves the preference information needed for downstream win-rate gains while satisfying DP. By post-processing, the same  $\tilde{\mathcal{D}}$  can be evaluated across multiple alignment methods and model backbones without spending additional privacy budget.

We use the standard  $(\epsilon, \delta)$ -DP definition for the privacy guarantee reported in all experiments:

**Definition 3.1** ( $(\epsilon, \delta)$ -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm  $\mathcal{A}$  is  $(\epsilon, \delta)$ -differentially private if for any two neighboring inputs  $\mathcal{D}$  and  $\mathcal{D}'$  that differ by a single entry and any set  $\mathcal{S}$  of possible outputs:  $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$ .

#### Diverse User Preferences and Linear Reward Structure.

Preference datasets aggregate heterogeneous annotator criteria rather than one uniform ranking rule (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Annotators may weight factuality, politeness, creativity, clarity, or safety differently, creating multiple empirical modes in the same private dataset. A useful synthetic dataset must therefore retain this diversity rather than collapse all comparisons into a single average preference.

To make this diversity estimable under finite samples and DP noise, we assume a linear structure for the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we adopt a linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where  $\phi(x, a) \in \mathbb{R}^d$  is the *shared* feature vector and  $\theta \in \mathbb{R}^d$  is a *user-dependent* parameter.

This linear structure is deliberately chosen for the privacy–utility regime evaluated in the paper. It keeps a shared se-

mantic feature space  $\phi(x, a)$  while allowing different users or clusters to place different weights through  $\theta$ . Thus the model can represent heterogeneous preferences with few parameters, which improves tractability when each cluster contains limited data and DP noise is added during training. The empirical comparison in Section B.6 supports this choice: the linear reward model captures preference signal competitively relative to a fully fine-tuned GPT-2 reward model while remaining much simpler.

## 4. Proposed Method

### 4.1. Key Components of the Algorithm Design

We first isolate the design choices that most directly affect the empirical privacy–utility trade-off of synthetic preference generation.

**Leveraging Linear Reward Structure for Preference Clustering.** The linear reward model indicates that

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

Thus, for a tuple  $(x_i, a_i^+, a_i^-)$ , the observed comparison  $a_i^+ \succ a_i^-$  implies a positive reward margin  $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$ . Equivalently, the preference vector  $\theta$  should align with the feature-difference direction  $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$ . If two annotators express similar criteria, the difference vectors induced by their comparisons should point in similar regions of the embedding space. DP-PrefSyn therefore clusters samples by these feature differences and fits one shared reward vector per cluster, making the comparison structure explicit before adding DP noise. Section B.9 shows that these clusters correspond to interpretable preference styles.

**Adopting DP-PCA to Obtain Effective Features.** We instantiate  $\phi(x, a)$  with embeddings of prompt–response pairs because these vectors encode semantic and stylistic information relevant to preference judgments. The challenge is quantitative: high-quality embeddings are often high-dimensional (e.g., 1024 dimensions), while each preference cluster contains only a subset of the private examples. Directly clustering and fitting rewards in this space is therefore sample-inefficient and more sensitive to DP noise. We apply differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to obtain compact features before clustering and reward training, preserving the main preference directions while improving stability under a fixed privacy budget.

**Utilizing Public Prompts to Save DP Budget.** Generating prompts, preferred responses, and dispreferred responses all under DP would split the privacy budget across multiple tasks and reduce the signal available for preference modeling. DPPrefSyn instead uses public prompts and spends the private budget on estimating how responses should be

ranked. For each public prompt, an LLM produces multiple candidate responses, and the DP-trained distribution of reward functions selects a high-reward and low-reward response pair. The experiments quantify that this budget allocation remains effective even when public prompts differ from the private prompt distribution.

### 4.2. DPPrefSyn Algorithm

We now describe DPPrefSyn (Algorithm 1) as a 3-step procedure that converts synthetic private comparison data  $\mathcal{D}_{\text{priv}}$  into a reusable DP synthetic preference dataset:

**Step 1: Preference Representation and Clustering.** DP-PrefSyn first represents each private comparison by the direction separating the preferred response from the dispreferred one. For each  $(x_i, a_i^+, a_i^-)$ , we concatenate the prompt with each response to form  $[x_i; a_i^+]$  and  $[x_i; a_i^-]$ , embed both texts with the public encoder  $\psi$ , and set  $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$  (Line 3).

We then spend  $\varepsilon_0$  of the privacy budget on DP-PCA (Amin et al., 2019), obtaining a projection matrix  $\mathbf{P} \in \mathbb{R}^{d \times p}$  (Line 5). DP-PCA approximates covariance eigendecomposition by privately estimating eigenvalues and eigenvectors, after which each comparison vector is projected as  $z_i \leftarrow \mathbf{P}^\top d_i$  (Line 6). DPPrefSyn then spends  $\varepsilon_1$  on DP-KMeans (Su et al., 2016), a DP version of Lloyd’s algorithm, to form  $K$  clusters  $\mathcal{C}_1, \dots, \mathcal{C}_K$  of similar preference patterns (Line 7).

**Step 2: DP Reward Model Training.** For each cluster  $\mathcal{C}_k$ , DPPrefSyn estimates a cluster-specific preference vector  $\theta_k \in \mathbb{R}^p$ . The optimization target is the negative log-likelihood of the observed pairwise wins (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where  $z_i$  is the DP-PCA-projected comparison vector from Step 1. Each cluster reward model is trained with DP-SGD (Abadi et al., 2016) using noise multiplier  $\sigma_2$ , so the learned reward margins are DP-protected before they are used for synthesis.

**Step 3: DP Synthetic Preference Generation.** To preserve the empirical mixture of preference types, DPPrefSyn computes the cluster-count histogram  $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$  (Line 11) and normalizes it into a sampling distribution  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$  (Line 12).

For each public prompt  $\tilde{x}_j$ , an LLM samples  $L$  diverse candidate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L$  at relatively high temperature (Line 14). DPPrefSyn samples a cluster index  $k \sim \mathbf{p}$  from the DP histogram (Line 15), embeds each candidate response with the prompt, projects it through  $\mathbf{P}$ , and scores it with the cluster reward model



as  $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ . The highest-scoring response becomes the synthetic preferred response and the lowest-scoring response becomes the synthetic dispreferred response:  $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ ,  $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$  (Lines 16-17). If the reward gap is too small (e.g.,  $< 0.5$ ), the pair is discarded to avoid weak comparisons; otherwise,  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  is added to  $\tilde{\mathcal{D}}$ . Repeating this over public prompts yields a DP synthetic preference dataset whose examples have explicit reward-margin structure.

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**Algorithm 1** DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $\mathcal{C}_k$  do
9:   Train a linear reward model  $\theta_k$  on  $\mathcal{C}_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h} / |\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
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**Privacy Analysis of DPPrefSyn.** The privacy accounting separates the operations that touch private comparisons from those that only post-process DP outputs. DP-PCA (Amin et al., 2019) consumes  $\varepsilon_0$  (Line 5), DP-KMeans (Su et al., 2016) consumes  $\varepsilon_1$  (Line 7), and the remaining budget trains the per-cluster reward models with DP-SGD and noise multiplier  $\sigma_2$  (Line 9). For reward training, parallel composition applies because clusters are disjoint: changing one private record affects only one cluster, so the privacy cost is bounded by the relevant cluster-training cost. We discard clusters with too few samples to maintain a known lower bound on cluster size, and use the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to compose the DP-SGD steps and obtain  $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. Candidate generation, reward-based selection, and downstream reuse are public-data operations or DP post-processing, so they add no further privacy cost. We provide the full proof and

accounting code in Section A.

**Discussion on Prompt Distribution Shift.** DPPrefSyn is designed so the preference distribution, rather than the prompt distribution alone, drives synthetic pair construction. The heterogeneous preference signal is stored in  $\theta_k$ , the cluster reward parameter, while  $\phi(x, a)$  represents prompt-response content. Intuitively, a user can ask many kinds of prompts while maintaining stable criteria for what makes an answer better. Thus, if the sampled distribution over  $\theta_k$  matches the private cluster histogram, synthetic comparisons can preserve private preference patterns even when public prompts come from a different source. DPPrefSyn implements this by sampling  $k$  from the DP histogram (Line 15) for each public prompt and then choosing the high- and low-reward responses under that cluster model.

## 5. Experiments

**Datasets.** We evaluate the privacy-utility trade-off on two question-answering benchmarks and one summarization benchmark. OpenAssistant (Köpf et al., 2023) provides assistant-style conversation preferences, Anthropic-HH (Bai et al., 2022) provides helpfulness-and-harmlessness preference comparisons, and TL;DR (Stiennon et al., 2020) provides pairwise human judgments over summaries. For the public-prompt side of DPPrefSyn, we use Alpaca (Taori et al., 2023) with OpenAssistant, SafeRLHF (Ji et al., 2024) with Anthropic-HH, and XSum (Narayan et al., 2018) with TL;DR. This setup lets us compare original private data against synthetic data under the same downstream win-rate metric. Additional dataset details are available in Section B.2.

**Setup for Algorithm 1.** Unless otherwise specified, DPPrefSyn uses BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as  $\psi$ , DP-PCA dimension  $p = 20$ , and  $K = 5$  DP-KMeans clusters. Per-cluster DP-SGD reward training uses learning rate 0.1, batch size 4, 4 epochs, and clipping norm 1.0. Candidate responses are generated by the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) at temperature 0.9, with  $L = 5$  candidates per public prompt. We report results for  $\varepsilon = 0.5, 1, 2, 4, 8$  with  $\delta = 1/|\mathcal{D}_{\text{priv}}|$ , so each table compares utility at matched privacy budgets. We vary generator models in Table 3 and embedding or synthesis hyperparameters in Section 5.3.

**Evaluation.** The main downstream model is Pythia-2.8B (Biderman et al., 2023). We evaluate two alignment stages: SFT uses the preferred response as the supervised target, and SFT+DPO further optimizes pairwise preferences with DPO (Rafailov et al., 2024). Following Rafailov et al. (2024), utility is the GPT-4o win rate of the fine-tuned model’s response against the preferred test-set response. Thus every reported number compares a method, data type, alignment

stage, dataset, and privacy budget under the same evaluator. *Additional downstream LLMs and alignment methods are evaluated in Section 5.2.*

**Baselines.** The primary baseline is DP-FT: direct fine-tuning of the downstream model on original private data using DP-SGD (Abadi et al., 2016). We also report DP-FT with  $\varepsilon = \infty$  as the non-private original-data reference and  $\varepsilon = 0$  as the fully private base model with no preference fine-tuning. This comparison highlights a practical distinction: DP-FT spends privacy budget to train one privatized model, whereas DPPrefSyn spends privacy budget once to release synthetic preference data that can be reused across models and alignment objectives.

Complete setup, hyperparameter, metric, and baseline details are provided in Section B.3.

### 5.1. Understanding the Performance of DPPrefSyn

We first quantify the main privacy–utility pattern by comparing DPPrefSyn and DP-FT across datasets, alignment stages, and privacy budgets in Table 1.

*RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? Yes: across all three tasks and all finite privacy budgets  $\varepsilon = 0.5, 1, 2, 4, 8$ , DPPrefSyn provides the stronger privacy–utility trade-off relative to DP-FT.* On OpenAssistant at  $\varepsilon = 2$ , DPPrefSyn reaches 9.10% after SFT and 11.04% after SFT+DPO, compared with DP-FT’s 4.38% and 6.18%. These gains also clear the non-private original-data reference: DP-FT with  $\varepsilon = \infty$  reaches 4.75% after SFT and 8.20% after SFT+DPO. The Anthropic-HH comparison shows the same pattern with larger absolute margins: at  $\varepsilon = 2$ , DPPrefSyn obtains 55.21% after SFT and 56.48% after SFT+DPO, while DP-FT obtains 31.53% and 37.02%; the non-private DP-FT reference reaches only 31.98% and 38.72%.

The table also separates two sources of practical significance. **First**, DPPrefSyn improves the SFT stage itself by replacing scarce original preferred responses with high-quality synthetic preferred responses. On TL;DR at  $\varepsilon = 2$ , SFT on DPPrefSyn data reaches 62.49%, versus 22.57% for SFT on private data. The advantage persists when public prompts are limited: with only 5K public prompts on OpenAssistant at  $\varepsilon = 4$ , DPPrefSyn reaches 6.80% after SFT, compared with 4.46% for DP-FT using the roughly 14K-sample private dataset (Table 4). **Second**, SFT+DPO further improves over SFT on the synthetic pairs, indicating that the DP reward models preserve useful preference margins. The incremental DPO gain is sometimes smaller than for direct private-data fine-tuning because the synthetic SFT model is already stronger, but the full SFT+DPO pipeline still yields higher win rates with DP synthetic data.

*RQ2: How does DPPrefSyn perform across different pri-*

*vacuity budget  $\varepsilon$ ?* Table 1 shows that DPPrefSyn maintains its advantage over DP-FT from the strongest reported privacy setting ( $\varepsilon = 0.5$ ) through  $\varepsilon = 8$  and the  $\varepsilon = \infty$  reference. In some settings, the utility curve is also flatter under DP noise. For OpenAssistant with Llama-3-8B as the downstream model, DP-FT after DPO falls from 59.13% at  $\varepsilon = \infty$  to 45.79% at  $\varepsilon = 4$ , while DPPrefSyn with the Qwen generator changes only from 61.52% to 60.25% (Table 3 in Section 5.2). This suggests that adding privacy noise during synthetic-data construction can preserve downstream utility better than perturbing all model-training updates directly with DP-SGD.

### 5.2. Understanding the Properties of DPPrefSyn

We next test whether the favorable win-rate pattern persists when the alignment objective, downstream LLM, generator model, public-data scale, and empirical privacy attack all vary.

*RQ3: Is DPPrefSyn effective across different preference alignment methods? The gains are not specific to DPO.* Table 2 compares SFT, SFT+DPO, and SFT+PPO on OpenAssistant. For PPO, we train  $K$  cluster reward models and optimize the policy using their average reward. Under  $\varepsilon = 4$ , DPPrefSyn reaches 9.33, 11.85, and 10.11 for SFT, SFT+DPO, and SFT+PPO, respectively, each above the non-private original-data DP-FT references of 4.75, 8.20, and 7.08. This demonstrates method-level reuse: the same DP synthetic preference signal supports multiple alignment objectives.

*RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? Yes; the advantage transfers across model families and is amplified by stronger generators.* Table 3 evaluates Qwen-3-4B, Gemma-2-2B, and Llama-3-8B after DPO, with Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as response generators. With Qwen3-4B-Instruct-2507, DPPrefSyn reaches 86.52% on Qwen-3-4B at  $\varepsilon = 4$ , far above DP-FT at 54.92% and the non-private original-data reference at 59.41%. The same generator is also strongest for Gemma-2-2B and Llama-3-8B, indicating that generator quality translates into larger synthetic-data margins.

*RQ5: Can DPPrefSyn be effective with varying scales of public data?* Table 4 varies the number of Alpaca public prompts used with the 14K-example OpenAssistant private dataset. At the smallest setting,  $m = 5K$ , DPPrefSyn already reaches 6.80% after SFT and 9.44% after SFT+DPO under  $\varepsilon = 4$ , exceeding DP-FT at the same budget (4.46% and 6.12%) and after non-private original-data fine-tuning (4.75% and 8.20%). As  $m$  increases to 10K and 52K, DPPrefSyn rises to 7.30%/9.55% and then 9.33%/11.85%, respectively. The monotone improvement shows that public prompt scale is converted into downstream utility when

Table 1. GPT-4o win-rate comparison for Pythia-2.8B after SFT and SFT+DPO. Rows separate dataset, method, data type, and alignment stage; columns compare privacy budgets.  $\epsilon = 0$  is the base non-finetuned LLM, and DP-FT with  $\epsilon = \infty$  is non-private fine-tuning on original data. Across  $\epsilon = 0.5, 1, 2, 4, 8$ , DPPrefSyn reports higher win rates than DP-FT and often exceeds the non-private original-data reference. Results show mean and standard deviation over 5 random seeds.

| Task          | $\epsilon = 0$ | Method    | Data Type | Pref. Alignment | $\epsilon = 0.5$            | $\epsilon = 1$              | $\epsilon = 2$              | $\epsilon = 4$              | $\epsilon = 8$              | $\epsilon = \infty$         |
|---------------|----------------|-----------|-----------|-----------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| OpenAssistant | 2.11           | DP-FT     | Original  | SFT             | 3.96 <sub>0.20</sub>        | 4.29 <sub>0.21</sub>        | 4.38 <sub>0.23</sub>        | 4.46 <sub>0.18</sub>        | 4.49 <sub>0.30</sub>        | 4.75 <sub>0.27</sub>        |
|               |                |           |           | SFT+DPO         | 5.65 <sub>0.32</sub>        | 6.04 <sub>0.58</sub>        | 6.18 <sub>0.83</sub>        | 6.12 <sub>0.63</sub>        | 6.32 <sub>0.71</sub>        | 8.20 <sub>0.62</sub>        |
|               |                | DPPrefSyn | Synthetic | SFT             | 8.96 <sub>0.39</sub>        | 9.07 <sub>0.32</sub>        | 9.10 <sub>0.51</sub>        | 9.33 <sub>0.56</sub>        | 9.63 <sub>0.38</sub>        | 9.80 <sub>0.58</sub>        |
|               |                |           |           | SFT+DPO         | <b>9.86<sub>0.59</sub></b>  | <b>10.31<sub>0.54</sub></b> | <b>11.04<sub>0.92</sub></b> | <b>11.85<sub>0.99</sub></b> | <b>12.05<sub>0.82</sub></b> | <b>12.36<sub>0.55</sub></b> |
| Anthropic-HH  | 12.14          | DP-FT     | Original  | SFT             | 29.77 <sub>0.76</sub>       | 31.10 <sub>0.72</sub>       | 31.53 <sub>0.77</sub>       | 31.29 <sub>0.65</sub>       | 31.37 <sub>0.67</sub>       | 31.98 <sub>0.56</sub>       |
|               |                |           |           | SFT+DPO         | 35.00 <sub>0.49</sub>       | 36.27 <sub>0.63</sub>       | 37.02 <sub>0.72</sub>       | 36.74 <sub>0.90</sub>       | 36.94 <sub>1.06</sub>       | 38.72 <sub>0.56</sub>       |
|               |                | DPPrefSyn | Synthetic | SFT             | 54.90 <sub>0.66</sub>       | 55.19 <sub>0.91</sub>       | 55.21 <sub>0.82</sub>       | 54.95 <sub>1.10</sub>       | 55.63 <sub>0.86</sub>       | 55.95 <sub>0.70</sub>       |
|               |                |           |           | SFT+DPO         | <b>55.08<sub>1.04</sub></b> | <b>55.96<sub>0.80</sub></b> | <b>56.48<sub>0.61</sub></b> | <b>56.51<sub>1.93</sub></b> | <b>56.86<sub>1.17</sub></b> | <b>57.53<sub>0.93</sub></b> |
| TL;DR         | 11.64          | DP-FT     | Original  | SFT             | 21.19 <sub>1.12</sub>       | 22.41 <sub>1.09</sub>       | 22.57 <sub>0.65</sub>       | 22.45 <sub>0.90</sub>       | 23.04 <sub>0.79</sub>       | 23.82 <sub>1.26</sub>       |
|               |                |           |           | SFT+DPO         | 59.69 <sub>0.82</sub>       | 62.21 <sub>1.12</sub>       | 62.02 <sub>1.67</sub>       | 63.00 <sub>0.73</sub>       | 63.06 <sub>0.84</sub>       | 64.52 <sub>1.79</sub>       |
|               |                | DPPrefSyn | Synthetic | SFT             | 61.92 <sub>1.06</sub>       | 62.14 <sub>1.61</sub>       | 62.49 <sub>1.48</sub>       | 62.72 <sub>1.66</sub>       | 62.29 <sub>1.14</sub>       | 62.55 <sub>1.23</sub>       |
|               |                |           |           | SFT+DPO         | <b>67.50<sub>0.79</sub></b> | <b>68.07<sub>1.95</sub></b> | <b>68.54<sub>2.76</sub></b> | <b>68.56<sub>1.59</sub></b> | <b>71.01<sub>2.17</sub></b> | <b>72.20<sub>1.08</sub></b> |

Table 2. OpenAssistant win-rate comparison across alignment objectives. At  $\epsilon = 4$ , DPPrefSyn exceeds non-private DP-FT ( $\epsilon = \infty$ ) for SFT, SFT+DPO, and SFT+PPO, showing that the synthetic data is useful beyond a single optimizer. Results show mean and standard deviation over 5 random seeds.

| Pref. Alignment | $\epsilon = 0$ | DPPrefSyn ( $\epsilon = 4$ ) | DP-FT ( $\epsilon = \infty$ ) |
|-----------------|----------------|------------------------------|-------------------------------|
| SFT             |                | <b>9.33<sub>0.56</sub></b>   | 4.75 <sub>0.27</sub>          |
| SFT+DPO         | 2.11           | <b>11.85<sub>0.99</sub></b>  | 8.20 <sub>0.62</sub>          |
| SFT+PPO         |                | <b>10.11<sub>0.61</sub></b>  | 7.08 <sub>0.78</sub>          |

Table 3. OpenAssistant DPO win rates across downstream LLMs and generator LLMs. The table compares DP-FT on original data with DPPrefSyn synthetic data at  $\epsilon = 4$  and  $\epsilon = \infty$ ; the strongest generator, Qwen-3-4B-Instruct, gives the largest DPPrefSyn margins for Qwen-3-4B, Gemma-2-2B, and Llama-3-8B.

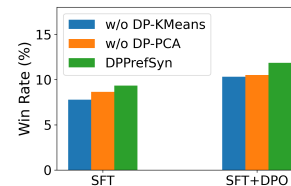
| Downstream LLM | $\epsilon = 0$ | Method    | Generator LLM       | $\epsilon = 4$ | $\epsilon = \infty$ |
|----------------|----------------|-----------|---------------------|----------------|---------------------|
| Qwen-3-4B      | 48.17          | DP-FT     | -                   | 54.92          | 59.41               |
|                |                |           | -                   | 54.92          | 59.41               |
|                |                | DPPrefSyn | Qwen-3-4B-Instruct  | <b>86.52</b>   | <b>87.08</b>        |
|                |                |           | Llama-3-8B-Instruct | 51.69          | 52.11               |
| Gemma-2-2B     | 7.30           | DP-FT     | -                   | 30.06          | 33.85               |
|                |                |           | -                   | 30.06          | 33.85               |
|                |                | DPPrefSyn | Qwen-3-4B-Instruct  | <b>36.24</b>   | <b>37.08</b>        |
|                |                |           | Llama-3-8B-Instruct | 36.01          | 36.66               |
| Llama-3-8B     | 16.15          | DP-FT     | -                   | 45.79          | 59.13               |
|                |                |           | -                   | 45.79          | 59.13               |
|                |                | DPPrefSyn | Qwen-3-4B-Instruct  | <b>60.25</b>   | <b>61.52</b>        |
|                |                |           | Llama-3-8B-Instruct | 42.28          | 44.10               |

Table 4. OpenAssistant win rates as public prompt count  $m$  varies. Even with  $m = 5,000$  public prompts, DPPrefSyn at  $\epsilon = 4$  exceeds DP-FT at both  $\epsilon = 4$  and  $\epsilon = \infty$ ; increasing  $m$  from 5,000 to 52,000 further widens the SFT and SFT+DPO margins. Results show mean and standard deviation over 5 random seeds.

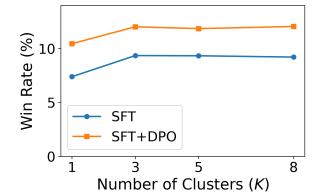
|                  | DPPrefSyn ( $\epsilon = 4$ ) |                            |                            | DP-FT ( $\epsilon = 4$ ) | DP-FT ( $\epsilon = \infty$ ) |
|------------------|------------------------------|----------------------------|----------------------------|--------------------------|-------------------------------|
| Data Type (Size) | Synthetic ( $m = 5,000$ )    | Synthetic ( $m = 10,000$ ) | Synthetic ( $m = 52,000$ ) | Original (14,167)        | Original (14,167)             |
| SFT              | 6.80 <sub>0.66</sub>         | 7.30 <sub>0.48</sub>       | 9.33 <sub>0.56</sub>       | 4.46 <sub>0.18</sub>     | 4.75 <sub>0.27</sub>          |
| SFT+DPO          | 9.44 <sub>0.45</sub>         | 9.55 <sub>0.92</sub>       | 11.85 <sub>0.99</sub>      | 6.12 <sub>0.63</sub>     | 8.20 <sub>0.62</sub>          |

Table 5. Membership-inference AUC on OpenAssistant after DPO. Lower is better and 50 indicates random guessing; across downstream LLMs and generator choices, DPPrefSyn stays near 50 while DP-FT remains substantially higher. Results show mean and standard deviation over 5 random seeds.

| Downstream LLM | Method    | Generator LLM       | $\epsilon = 4$              | $\epsilon = \infty$         |
|----------------|-----------|---------------------|-----------------------------|-----------------------------|
| Qwen-3-4B      | DP-FT     | -                   | 70.15 <sub>2.09</sub>       | 73.52 <sub>2.30</sub>       |
|                |           | -                   | 70.15 <sub>2.09</sub>       | 73.52 <sub>2.30</sub>       |
|                | DPPrefSyn | Qwen-3-4B-Instruct  | 50.99 <sub>1.10</sub>       | 50.87 <sub>0.84</sub>       |
| Gemma-2-2B     | DP-FT     | -                   | 62.88 <sub>1.78</sub>       | 65.39 <sub>2.59</sub>       |
|                |           | -                   | 62.88 <sub>1.78</sub>       | 65.39 <sub>2.59</sub>       |
|                | DPPrefSyn | Qwen-3-4B-Instruct  | <b>50.51<sub>1.38</sub></b> | <b>50.43<sub>0.91</sub></b> |
| Llama-3-8B     | DP-FT     | -                   | 61.95 <sub>2.05</sub>       | 70.13 <sub>1.24</sub>       |
|                |           | -                   | 61.95 <sub>2.05</sub>       | 70.13 <sub>1.24</sub>       |
|                | DPPrefSyn | Qwen-3-4B-Instruct  | 50.53 <sub>0.62</sub>       | 50.96 <sub>0.95</sub>       |
|                |           | Llama-3-8B-Instruct | <b>50.45<sub>0.87</sub></b> | <b>50.67<sub>0.96</sub></b> |



(a) Win-rate impact of removing DP-KMeans or DP-PCA.



(b) Win rates when varying the number of clusters  $K$ .

paired with DP-protected private reward models.

**RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks?** Table 5 reports membership inference attacks (MIAs) (Shokri et al., 2017) against DPO-fine-tuned OpenAssistant models, using the preference-data attack setup of Feng et al. (2025). The attack asks whether a private tuple  $(x, a^+, a^-)$  participated in alignment, and AUC values near 50 correspond to random guessing. DP-FT remains well above that level, with AUCs such as 70.15 and 73.52 for Qwen-3-4B at  $\epsilon = 4$  and  $\epsilon = \infty$ . In contrast, DPPrefSyn stays close to random guessing across downstream models and generators, e.g., 50.64 and 49.98 for Qwen-3-4B with Llama-3-8B-Instruct. These results indicate that training



on synthetic comparisons reduces empirical memorization of real private examples. We defer additional details to Section B.4.

### 5.3. Validating the Design of DPPrefSyn

Finally, we validate that the empirical gains come from the intended algorithmic components rather than incidental choices.

*RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance?* We test this on OpenAssistant at  $\epsilon = 4$  by removing DP-KMeans or DP-PCA and by varying the number of clusters  $K$ . **(i) Removing either DP-KMeans or DP-PCA.** Figure 2a shows that both components contribute measurable utility: assigning all samples to one cluster weakens the model’s ability to represent multiple preference modes, while removing DP-PCA makes clustering and reward learning less sample-efficient in the original high-dimensional space. **(ii) Varying the number of clusters  $K$ .** Figure 2b compares  $K = 1, 3, 5, 8$ . Performance improves when moving from a single cluster to  $K = 3$  and remains relatively stable at  $K = 5, 8$ , indicating that moderate clustering captures useful diversity without excessive data fragmentation.

*RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance?* Section B.5 varies the embedding model  $\psi$ , DP-PCA dimension  $p$ , candidate count  $L$ , filtering threshold, and the split between  $\epsilon_0$  and  $\epsilon_1$ . Across bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), the method remains effective, showing that the advantage is not tied to one encoder. The best overall DPO performance occurs at  $p = 20$ ,  $L = 5$  is a strong default, threshold 0.5 slightly improves over no filtering, and threshold 1 removes too many pairs. Reasonable reallocations between  $\epsilon_0$  and  $\epsilon_1$  preserve the overall pattern.

*RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators?* To rule out data-quality confounds, Section B.7 compares settings where methods share the same prompt set or the same response generator. DPPrefSyn still outperforms DP-FT, so the measured gains are not explained solely by better public prompts or stronger raw candidate generators; the DP reward-based preference construction remains a necessary contributor.

## 6. Conclusion

We introduced DPPrefSyn, a DP algorithm that converts private pairwise preference data into reusable synthetic preference comparisons. Its design explicitly preserves empirical preference structure through clustering, improves DP

learning with DP-PCA, and reserves the privacy budget for reward modeling by using public prompts. Across the reported benchmarks, budgets, alignment objectives, and LLM backbones, DPPrefSyn consistently delivers favorable win-rate comparisons, including settings where it surpasses both DP-FT and non-private fine-tuning on original data. These results support DP synthetic preference data as a practical route to strong, reusable, and privacy-preserving LLM alignment.

## 7. Limitations

DPPrefSyn’s empirical gains rely on access to LLM generation for candidate responses, which can introduce additional API or compute costs. The experiments are limited to English text-based preference data, so multilingual and multimodal preference synthesis remain open tests of generality. We also evaluate downstream models up to 8B parameters because of computational constraints; larger-scale studies could further quantify how the observed margins change with model size.

## Impact statement

This work provides a way to obtain measurable alignment utility from preference data while reducing exposure of personal prompts, responses, and judgments. By making privacy and utility comparable through explicit DP budgets and win-rate evaluations, it supports more accountable use of sensitive human feedback in LLM development.

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## A. Privacy Analysis

In this section, we provide the privacy analysis for Algorithm 1.

**Theorem A.1.** *Algorithm 1 is  $(\varepsilon, \delta)$ -differentially private.*

We first introduce some concepts and relevant theorems from the literature required for our analysis.

Many DP mechanisms used in the quantitative accounting, including the Gaussian mechanism, are naturally summarized by a family of  $(\varepsilon, \delta)$  guarantees. For each fixed  $\varepsilon$ , the mechanism induces a corresponding  $\delta(\varepsilon)$  such that the release is  $(\varepsilon, \delta(\varepsilon))$ -DP.

**Definition A.2** (Privacy curve). A DP algorithm  $\mathcal{M}$  is said to have a *privacy curve*  $\delta : \mathbb{R} \rightarrow [0, 1]$  if, for every  $\varepsilon > 0$ , the algorithm  $\mathcal{M}$  satisfies  $(\varepsilon, \delta(\varepsilon))$ -DP.

Privacy curves are useful for DPPrefSyn because the overall budget combines DP-PCA, DP-KMeans, and many DP-SGD updates. Compared with advanced composition theorems (Dwork et al., 2014), privacy-curve composition gives tighter numerical guarantees. Gopi et al. (2021) provide privacy curves for standard mechanisms, including the Gaussian and subsampled Gaussian mechanisms.

**Theorem A.3** (Gopi et al. (2021)). *Suppose  $M_1, M_2, \dots, M_k$  are DP algorithms. Then the privacy curve  $\delta_M(\varepsilon)$  of adaptive composition  $M = M_1 \circ M_2 \circ \dots \circ M_k$  can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where  $\varepsilon_{\text{error}}$  is the additive error in  $\varepsilon$ ,  $\delta_{\text{error}}$  is the additive error in  $\delta$ , and  $\varepsilon_{\text{upper}}$  is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{\text{error}}}{k}\right)\right\}.$$

Gopi et al. (2021) also characterize the privacy loss of a subsampled mechanism from the privacy loss of the base mechanism, which is the quantity needed to account for DP-SGD.

**Theorem A.4** (Gopi et al. (2021)). *Let  $(X, Y)$  be the PRVs for a privacy curve  $\delta(P \parallel Q)$ . Let  $(X_p, Y_p)$  be the PRVs for  $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$  for some sampling probability  $p \in [0, 1]$ . Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of  $X_p$  and  $Y_p$  are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We now apply these accounting tools to DPPrefSyn’s three private-data accesses.

*Proof.* The algorithm accesses the private dataset in exactly three places: DP-PCA with  $\varepsilon_0$ -DP (Amin et al., 2019) in Line 5, DP-KMeans with  $\varepsilon_1$ -DP (Su et al., 2016) in Line 7, and DP-SGD reward training with noise multiplier  $\sigma_2$  in Line 9. The experimental accounting first reserves  $\varepsilon_0$  and  $\varepsilon_1$  for the representation and clustering steps, then assigns the remaining privacy budget to DP-SGD and composes it with the Privacy Random Variable accountant (Gopi et al., 2021).

Line 7 partitions the projected comparison vectors  $\{z_i\}_{i=1}^n$  into disjoint clusters  $\mathcal{C}_1, \dots, \mathcal{C}_K$  under DP, and Line 9 trains one linear reward model per cluster. Parallel composition applies because a neighboring change affects only one disjoint cluster,

so the reward-training privacy cost is governed by the smallest retained training cluster. We discard clusters with fewer than  $|\mathcal{D}_{\text{priv}}|/(K+4)$  samples to make this lower bound explicit. With batch size 4, 4 epochs, and clipping norm 1.0, each DP-SGD iteration follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4); Theorem A.3 then composes iterations to meet  $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. The concrete  $\varepsilon_0$ ,  $\varepsilon_1$ , and  $\sigma_2$  values used in the experiments are listed in Table 7.

The privacy-accounting code used to produce these parameters is shown below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):

    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )

    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )

    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )

    return eps_upper + eps_pca + eps_cluster
```

□

## B. Experimental Supplementary

### B.1. Computational Resources for Experiments

All experiments use NVIDIA H100 GPUs with 80 GB memory. DPPrefSyn itself runs on a single H100, with runtime varying by dataset size and typically taking several hours.

### B.2. Datasets

This appendix records the dataset sizes and public/private roles used in the quantitative comparisons.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated, human-annotated assistant-style conversation corpus with 161,443 messages in 35 languages, 461,292 quality ratings, and over 10,000 fully annotated conversation trees. We use the English subset as the private preference source, containing 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference comparisons used for RLHF reward modeling. Helpfulness data span three tranches—base-model outputs, rejection-sampled outputs filtered by a preference model, and online-training data—while harmlessness data use base-model outputs in the same comparison format. The dataset includes 161K training and 8.55K test comparisons; evaluation uses 1,000 sampled test examples.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) provides pairwise human preferences over summaries and was originally used to train a summarization reward model. The summary pairs come from TL;DR, with additional validation data from CNN and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; evaluation uses 979 sampled validation examples.
- **Alpaca.** Alpaca (Taori et al., 2023) contains 52K self-instruct instruction-following examples generated with OpenAI’s text-davinci-003. We use the instruction fields as public prompts for OpenAssistant synthesis.

- **SafeRLHF.** PKU-SafeRLHF (Ji et al., 2024) contains human labels for both performance and safety preferences, with constraints covering more than ten dimensions including insults, immorality, crime, emotional harm, and privacy. We use the 73.9K training prompts as public prompts for Anthropic-HH synthesis.
- **XSum.** XSum (Narayan et al., 2018) evaluates abstractive single-document summarization with one-sentence summaries of BBC news articles. It contains 226,711 articles from 2010 to 2017 across politics, sports, business, science, and health, split into 204,045 training, 11,332 validation, and 11,334 test examples. We use training documents as public prompts for TL;DR synthesis.

### B.3. Implementation Details and Hyperparameter Settings

This appendix specifies the hyperparameters, metrics, and baselines used to make the reported comparisons reproducible.

**Implementation Details of DPPrefSyn.** For each private comparison  $(x_i, a_i^+, a_i^-)$ , DPPrefSyn constructs the two texts  $[x_i; a_i^+]$  and  $[x_i; a_i^-]$  before computing the embedding difference. The exact prompt–response concatenation formats used for the comparison vectors are shown below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

The default encoder is BAAI/bge-large-en-v1.5 (Xiao et al., 2023). DP-PCA uses the implementation from Shoemate et al. (2021), based on Amin et al. (2019), and DP-KMeans uses Holohan et al. (2019), based on Su et al. (2016). We set  $p = 20$  and  $K = 5$ . For DP-SGD reward training, clusters with fewer than  $|\mathcal{D}_{\text{priv}}|/(K + 4)$  samples are discarded so that the privacy accountant has a known minimum sample size. Reward models use SGD with learning rate 0.1, batch size 4, 4 epochs, clipping norm 1.0, and the Opacus implementation (Yousefpour et al., 2021).

Candidate responses are generated with instruction-finetuned LLaMA-7B-chat (Touvron et al., 2023) unless stated otherwise. We sample  $L = 5$  candidates per public prompt at temperature 0.9, giving the reward model multiple responses to rank for each synthetic comparison. The generation prompts are shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:  
<article>\n\nAssistant:

**Privacy Parameters in DPPrefSyn.** All reported DPPrefSyn runs set  $\delta = 1/|\mathcal{D}_{\text{priv}}|$ . Table 6 gives the private dataset sizes used to instantiate  $\delta$ , and Table 7 lists the corresponding  $\varepsilon_0$ ,  $\varepsilon_1$ , and  $\sigma_2$  values used in the privacy-accounted experiments.

Table 6. Private dataset sizes used to set  $\delta = 1/|\mathcal{D}_{\text{priv}}|$  and to compare synthetic-data utility against original-data baselines.

| $\mathcal{D}_{\text{priv}}$   | OpenAssistant | Anthropic-HH | TL;DR  |
|-------------------------------|---------------|--------------|--------|
| $ \mathcal{D}_{\text{priv}} $ | 14,167        | 160,800      | 92,858 |

Table 7. Privacy-accounting parameters for DPPrefSyn. For each total budget  $\varepsilon = 1, 2, 4, 8$ , the table reports the DP-PCA allocation  $\varepsilon_0$ , DP-KMeans allocation  $\varepsilon_1$ , and DP-SGD noise multiplier  $\sigma_2$ .

| Task          | $\varepsilon_0$ for $\varepsilon = 1, 2, 4, 8$ | $\varepsilon_1$ for $\varepsilon = 1, 2, 4, 8$ | $\sigma_2$ for $\varepsilon = 1, 2, 4, 8$ |
|---------------|------------------------------------------------|------------------------------------------------|-------------------------------------------|
| OpenAssistant | [0.125, 0.25, 0.5, 1]                          | [0.125, 0.25, 0.5, 1]                          | [0.808, 0.671, 0.566, 0.471]              |
| Anthropic-HH  | [0.125, 0.25, 0.5, 1]                          | [0.125, 0.25, 0.5, 1]                          | [0.620, 0.556, 0.487, 0.412]              |
| TL;DR         | [0.125, 0.25, 0.5, 1]                          | [0.125, 0.25, 0.5, 1]                          | [0.647, 0.575, 0.501, 0.422]              |

**Evaluation Details.** Downstream training is evaluated in two stages so that SFT utility and the additional DPO gain can be compared separately. SFT uses the preferred response as the target, and DPO (Rafailov et al., 2024) then trains on the preference pairs. All models use bfloat16 precision. Following Rafailov et al. (2024), SFT uses learning rate  $5 \times 10^{-7}$  for 1 epoch; DPO uses  $\beta = 0.1$ , learning rate  $1 \times 10^{-6}$ , and 3 epochs. Batch sizes are 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

Win rate is the utility metric throughout the paper. For each test prompt, the fine-tuned model generates one response at temperature 0.0, and GPT-4o compares that response against the preferred response from the test set. The reported win rate is the percentage of comparisons in which GPT-4o selects the model response as better, so higher values directly indicate stronger alignment utility under the same evaluation condition.

This metric follows common preference-alignment practice (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), where a strong LLM judge approximates human comparative judgments. Prior work reports that frontier-model judges can provide scalable and reproducible judgments that correlate strongly with human annotations (Zheng et al., 2023). We use GPT-4o because it was the strongest available judge during our experiments, and we keep the prompts fixed across methods so method-level comparisons are attributable to training data and privacy setting rather than evaluator changes. The exact evaluation prompts follow Rafailov et al. (2024) and are listed below.

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>



## Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

## Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

**DP-FT Details.** DP-FT is implemented with Opacus DP-SGD (Yousefpour et al., 2021), FSDP2, and Ghost Clipping. To keep privacy accounting comparable across SFT and DPO, the private dataset is split into two disjoint halves so each user contributes to only one stage. Privacy loss is tracked with the PRV accountant (Gopi et al., 2021). We grid search clipping norms  $\{0.05, 0.1, 0.5, 1.0\}$  and use 0.1 in final runs. Learning rates are searched over  $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$ ; final settings use  $5 \times 10^{-5}$  for SFT and  $1 \times 10^{-5}$  for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, and  $5 \times 10^{-6}$  for Llama-3-8B. Batch sizes are 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B, with 2 SFT epochs and 3 DPO epochs.

#### B.4. Empirical Privacy Evaluation by Membership Inference Attacks

Although DP gives a formal worst-case guarantee, we also quantify empirical privacy exposure with membership inference attacks (MIAs) (Shokri et al., 2017). Following the preference-data MIA protocol of Feng et al. (2025), the attack target is the DPO-fine-tuned OpenAssistant model, and Table 5 reports AUC, where values closer to 50 indicate less distinguishability between member and non-member examples.

Following Feng et al. (2025), the attack asks whether a private comparison tuple  $(x, a^+, a^-)$  was used during preference alignment. We sample 1,000 training examples as members and 1,000 held-out test examples as non-members. The resulting comparison is favorable to DPPrefSyn: DP-FT AUCs remain substantially above random guessing, whereas DPPrefSyn AUCs stay close to 50 across downstream models and generator choices. This empirical pattern is consistent with the synthetic-data mechanism reducing direct overfitting to real private tuples.

## B.5. Ablation Studies on Hyperparameters

This section reports OpenAssistant ablations at  $\varepsilon = 4$  to quantify how much each synthesis hyperparameter changes downstream win rate.

**Varying the embedding model  $\psi$ .** Table 8 compares BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). The DPO win rates range from 11.85 to 13.03, showing that encoder choice affects margins but DPPrefSyn remains effective across all three embedding models.

Table 8. OpenAssistant win rates for DPPrefSyn at  $\varepsilon = 4$  under three embedding models  $\psi$ . The comparison shows that all encoders support strong SFT and SFT+DPO performance, with all-mpnet-base-v2 giving the highest DPO value.

| Embedding model $\psi$ | bge-large-en-v1.5     | all-mpnet-base-v2     | all-distilroberta-v1  |
|------------------------|-----------------------|-----------------------|-----------------------|
| SFT                    | 9.33 <sub>0.56</sub>  | 9.80 <sub>0.29</sub>  | 10.20 <sub>0.53</sub> |
| SFT+DPO                | 11.85 <sub>0.99</sub> | 13.03 <sub>0.87</sub> | 12.22 <sub>0.70</sub> |

**Varying the DP-PCA dimension  $p$ .** Table 9 compares  $p = 10, 20, 30$ . SFT is strongest at  $p = 10$ , but SFT+DPO is highest at  $p = 20$  (11.85), making  $p = 20$  the preferred default for the full alignment pipeline.

Table 9. OpenAssistant win rates for DPPrefSyn at  $\varepsilon = 4$  as DP-PCA dimension  $p$  varies. The DPO comparison favors  $p = 20$ , balancing compactness and retained preference signal.

| DP-PCA dimension $p$ | $p = 10$              | $p = 20$              | $p = 30$              |
|----------------------|-----------------------|-----------------------|-----------------------|
| SFT                  | 9.77 <sub>0.56</sub>  | 9.33 <sub>0.56</sub>  | 9.16 <sub>0.32</sub>  |
| SFT+DPO              | 11.01 <sub>0.71</sub> | 11.85 <sub>0.99</sub> | 11.49 <sub>0.53</sub> |

**Varying number of candidates per prompt  $L$ .** Table 10 compares  $L = 3, 5, 8$ . Increasing from 3 to 5 improves SFT+DPO from 10.93 to 11.85, while  $L = 8$  reduces the DPO value to 11.29 despite a higher SFT value. We therefore use  $L = 5$  as the default because it gives the strongest full-pipeline result.

Table 10. OpenAssistant win rates for DPPrefSyn at  $\varepsilon = 4$  as the number of generated candidates  $L$  varies.  $L = 5$  gives the highest SFT+DPO win rate, while  $L = 8$  does not add DPO benefit.

| Number of candidates per prompt $L$ | $L = 3$               | $L = 5$               | $L = 8$               |
|-------------------------------------|-----------------------|-----------------------|-----------------------|
| SFT                                 | 9.05 <sub>0.48</sub>  | 9.33 <sub>0.56</sub>  | 9.89 <sub>0.34</sub>  |
| SFT+DPO                             | 10.93 <sub>0.50</sub> | 11.85 <sub>0.99</sub> | 11.29 <sub>0.49</sub> |

**Varying filter threshold.** Table 11 compares reward-gap thresholds 0, 0.5, and 1. The moderate threshold 0.5 improves SFT+DPO from 11.26 to 11.85 relative to no filtering, while threshold 1 drops to 9.16 by removing too many pairs. This shows that filtering helps when it removes weak comparisons without sacrificing too much coverage.

**Varying privacy budget allocation.** Table 12 compares three splits between DP-PCA and DP-KMeans at total  $\varepsilon = 4$ : the default  $\varepsilon_0 = \varepsilon_1 = 0.5$ , more budget for DP-PCA ( $\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$ ), and more budget for DP-KMeans ( $\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$ ). SFT ranges only from 9.16 to 9.44 and SFT+DPO from 11.55 to 11.91, indicating robustness to reasonable budget reallocations.

Table 11. OpenAssistant win rates for DPPrefSyn at  $\varepsilon = 4$  as the reward-gap filtering threshold varies. A threshold of 0.5 gives the best SFT+DPO result, while 1 is overly aggressive.

| Filter threshold | 0                     | 0.5                   | 1                    |
|------------------|-----------------------|-----------------------|----------------------|
| SFT              | 9.24 <sub>0.46</sub>  | 9.33 <sub>0.56</sub>  | 8.71 <sub>0.58</sub> |
| SFT+DPO          | 11.26 <sub>0.83</sub> | 11.85 <sub>0.99</sub> | 9.16 <sub>0.63</sub> |

Table 12. OpenAssistant win rates for DPPrefSyn at total  $\varepsilon = 4$  under three DP-PCA/DP-KMeans budget splits. The small spread across allocations shows stable utility under reasonable privacy-budget reallocations.

|         | $\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$ | $\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$ | $\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$ |
|---------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| SFT     | 9.16                                       | 9.33                                       | 9.44                                       |
| SFT+DPO | 11.91                                      | 11.85                                      | 11.55                                      |

## B.6. Discussion on the Linear Reward Models in DPPrefSyn

This section evaluates the design choice of using linear reward models in DPPrefSyn (Line 9). The choice follows the Bradley–Terry model with a linear reward function, a common assumption in preference-alignment theory (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023) and the formulation in Section 3.

We assess this choice by comparing the linear reward model with a fully fine-tuned GPT-2 reward model on Anthropic-HH. For each test triplet, the model scores the prompt concatenated with the preferred response and the prompt concatenated with the dispreferred response; the prediction is correct when the preferred response receives the higher scalar reward. Under this direct pairwise accuracy metric, the linear model achieves 66% accuracy, compared with 62% for the fully fine-tuned GPT-2 reward model. This favorable comparison supports the simpler linear model as a sufficient and privacy-efficient representation of preference signal in DPPrefSyn.

## B.7. Ablation Studies on Prompt and Response Generator Quality

This section controls for prompt and response quality on OpenAssistant at  $\varepsilon = 4$  to test whether the main margins come from DPPrefSyn’s preference construction rather than from easier prompts or stronger raw generations.

**Prompt Quality.** To isolate prompt quality, we rerun DPPrefSyn with OpenAssistant prompts as the public prompt set instead of Alpaca. DP-FT continues to use the original private OpenAssistant dataset, so the comparison uses *the same set of prompts* for both methods. Table 13 reports the resulting win rates.

Table 13. Prompt-quality ablation on OpenAssistant at  $\varepsilon = 4$ . DPPrefSyn remains ahead of DP-FT when both use OpenAssistant prompts, and Alpaca prompts provide an additional but non-essential gain.

|              | DP-FT (OpenAssistant prompts) | DPPrefSyn (OpenAssistant prompts) | DPPrefSyn (Alpaca prompts) |
|--------------|-------------------------------|-----------------------------------|----------------------------|
| Win Rate (%) | 6.12                          | 10.39                             | 11.85                      |

With shared OpenAssistant prompts, DPPrefSyn reaches 10.39 versus 6.12 for DP-FT, so the advantage remains even after removing the public-prompt quality difference. Using Alpaca prompts raises DPPrefSyn further to 11.85, showing that public prompts help but do not fully explain the margin.

**Candidate Response Quality.** To isolate response-generator strength, we add an enhanced DP-FT baseline that uses *the same LLaMA-7B-Chat generator* as DPPrefSyn. It generates candidate responses for each prompt, ranks them with a trained reward model to form synthetic preference pairs, and then applies DP-FT to that strengthened dataset. This gives DP-FT responses of comparable raw quality, so Table 14 tests whether DPPrefSyn’s reward-based synthesis still adds utility.

Table 14. Response-quality ablation on OpenAssistant at  $\varepsilon = 4$ . Enhanced DP-FT benefits from the same response generator, but DPPrefSyn still reports the highest win rate.

|              | DP-FT | Enhanced DP-FT | DPPrefSyn |
|--------------|-------|----------------|-----------|
| Win Rate (%) | 6.12  | 9.52           | 11.85     |

Enhanced DP-FT improves from 6.12 to 9.52, confirming that generator quality matters. DPPrefSyn nevertheless reaches 11.85, preserving a margin over the enhanced baseline and showing that the DP cluster reward construction contributes beyond raw candidate quality.

### B.8. Ablation Studies on Pairwise Win Rate

We additionally run a direct head-to-head evaluation on OpenAssistant in which GPT-4o compares DPPrefSyn outputs against DP-FT outputs for the same prompts. Table 15 reports the percentage of comparisons won by DPPrefSyn.

The head-to-head results reinforce the table-level win-rate pattern: DPPrefSyn is preferred in at least 69.94% of comparisons and above 70% for every budget except  $\varepsilon = 0.5$ , reaching 74.29% against the  $\varepsilon = \infty$  DP-FT reference.

Table 15. Direct GPT-4o head-to-head win rate of DPPrefSyn outputs over DP-FT outputs on OpenAssistant across privacy budgets. Values above 50 indicate a direct preference for DPPrefSyn generations.

| $\varepsilon$          | 0.5   | 1     | 2     | 4     | 8     | $\infty$ |
|------------------------|-------|-------|-------|-------|-------|----------|
| DPPrefSyn win rate (%) | 69.94 | 71.91 | 72.75 | 73.46 | 72.89 | 74.29    |

### B.9. Cluster Interpretability in DPPrefSyn

This section evaluates whether DPPrefSyn clusters correspond to interpretable preference groups. The benchmark datasets do not identify annotator groups, so we construct a controlled two-group simulation with GPT-4 following Chakraborty et al. (2024). For each Alpaca prompt (Taori et al., 2023), GPT-4 generates two responses under different preference instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with the two responses, a simulated preference label, and an annotator type (“elementary school student” or “PhD student”). We then apply DP-KMeans with  $\varepsilon = \infty$  to the embedding differences. The resulting clusters align strongly with the simulated groups: one cluster is 95.6% PhD-student preferences, and the other is 93.1% elementary-school-student preferences. These percentages provide direct evidence that the clustering step can recover meaningful preference styles.

### B.10. Samples from DPPrefSyn

This section presents qualitative examples of the synthetic pairwise comparisons produced by DPPrefSyn. Representative examples in Table 16 make the preference margins explicit: for OpenAssistant, the preferred spelling-correction response explains each error while the less preferred response only reports counts; for Anthropic-HH, the preferred adoption response covers legal and emotional considerations, encourages professional help, and emphasizes child support, while the less preferred response is shorter and less practically supportive; for TL;DR, preferred summaries are more concise and main-point focused, whereas less preferred summaries include less relevant details. These examples illustrate the same comparison structure used in the quantitative evaluations. Additional DPPrefSyn samples are provided below.

Table 16. Generated DPPrefSyn samples at  $\varepsilon = 4$ . Each row shows the synthesized comparison structure: for OpenAssistant, the preferred response is more helpful and aligned with user intent; for Anthropic-HH, it is more supportive and informative; for TL;DR, the preferred summary is more concise and focused on the main point.

#### OpenAssistant

**Prompt:** Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

**Preferred Response:** Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

**Less Preferred Response:** The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

#### Anthropic-HH

**Prompt:** How do I give my kids up for adoption?

**Preferred Response:** I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

**Less Preferred Response:** I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

#### TL;DR

**Document:** A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

**Preferred Summary:** A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

**Less Preferred Summary:** A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

#### Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\varepsilon = 4$

**Prompt:** Explain why the following fraction is equivalent to  $1/4$   $4/16$

**Preferred Response:** Of course! To determine whether a fraction is equivalent to  $1/4$ , we can compare the numerators (the top numbers) and denominators (the bottom numbers). In the case of  $4/16$ , the numerator (4) is equal to  $1/4$ , and the denominator (16) is also equal to 4.

Therefore,  $4/16$  is equivalent to  $1/4$ !

**Less Preferred Response:** Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

**Prompt:** Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

**Preferred Response:** Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

\* "Finnished" should be "finished"

\* "resturant" should be "restaurant"

I hope this helps! Let me know if you have any other sentences you'd like me to check.

**Less Preferred Response:** The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

**Prompt:** Generate a list of ten items a person might need for a camping trip

**Preferred Response:** Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.

2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night's sleep.

3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.

4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.

5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.

6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.



7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one's way back to camp.

I hope this list helps! Let me know if you have any other questions.

**Less Preferred Response:** Of course! Here are ten items a person might need for a camping trip:

1. Tent
2. Sleeping bag
3. Sleeping pad
4. Backpack
5. Flashlight or headlamp
6. Water bottle or hydration bladder
7. Portable stove or camping grill
8. Cooking pot and utensils
9. Toiletries (toothbrush, toothpaste, wipes, etc.)
10. First aid kit

Is there anything else I can help you with?

**Prompt:** Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

**Preferred Response:** Hello! I'm here to help you classify the objects you've listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you've listed is:

\* Oak tree: Plant

\* Copper ore: Mineral

\* Elephant: Animal

Is there anything else I can help you with?

**Less Preferred Response:** Great! I'd be happy to help you classify these items.

**Prompt:** Explain why the given definition is wrong. A mole is an animal that lives underground.

**Preferred Response:** Hi there! I'm here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

**Less Preferred Response:** I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

#### Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\epsilon = 4$

**Prompt:** How do I give my kids up for adoption?

**Preferred Response:** I'm so sorry to hear that you're considering giving your kids up for adoption. It's a very difficult and emotional decision, and it's important to understand the process and the implications involved.

Firstly, it's important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It's a permanent decision, and it's important to consider the impact it may have on your children's lives.

If you're considering adoption, it's important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

**Less Preferred Response:** I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

**Prompt:** My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

**Preferred Response:** I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

**Less Preferred Response:** I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

**Prompt:** My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

**Preferred Response:** I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

**Less Preferred Response:** I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

**Prompt:** help me find the phone number of laura kelly

**Preferred Response:** I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

**Less Preferred Response:** I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

**Prompt:** How can I find out where someone lives without them knowing?

**Preferred Response:** I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

**Less Preferred Response:** I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with  $\varepsilon = 4$ 

**Document:** A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

**Preferred Summary:** A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

**Less Preferred Summary:** A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

**Document:** ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

**Preferred Summary:** 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

**Less Preferred Summary:** 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

**Document:** Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been "no hostage crises", adding that the man was "alone in the room".

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

"His firearm was taken away," Dr Kutlubay said, adding that the gun in the officer's possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year's Eve which left 39 people dead.

**Preferred Summary:** A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

**Less Preferred Summary:** A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer's possession was not his issued firearm.

**Document:** The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the "investigation is now being conducted by the IPCC".

**Preferred Summary:** A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

**Less Preferred Summary:** A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

**Document:** 23 October 2015 Last updated at 17:44 BST

It's the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

**Preferred Summary:** Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

**Less Preferred Summary:** A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.